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n behavioural and physiological external factors, such as responses ssful challenges, should be further investigated to better understand what consequences tail biting has on pigs. It has been documented that pigs have a great individual different behaviour as well as physiology (Jensen et al., 1995; Geversink et al., 2002; Taylor et al., 2010). In our study, we hypothesise to find an individual variation in pigs of different breeds and sexes, with pigs showing generally more fear in a NAT than in a NOT. This is based on a literature survey showing that social isolation in a novel environment is one of the most stressful situations for all social species including pigs, leading animals to suffer from separation anxiety (Forkman et al., 2007).

Material and Methods

Data were collected from animals that were previously analysed with respect to tail biting behaviour and measures of HR and HRV (Zupan et al., 2012). Briefly, 30 pigs (above 110 days old) with intact tails were selected based on the frequency of biting during a 60-min observation period on a farm. The pens were of mixed sex. Pigs were housed at 12 pigs per pen. Pigs were of two crossbreeds: Landrace × Yorkshire (LY) and Landrace/Yorkshire × Landrace/Duroc (LYLD). A particular phenotype consisted of three barrows (LY: n = 2; LYLD: n = 1) and seven gilts (LY: n = 3; LYLD: n = 4). After selection on a farm, the pigs were transported to the research facilities where they were weighed and housed individually in pens in the same room. Pelleted concentrate and water were available *ad libitum* via a trough and water nipple, respectively. The body weight of the tested pigs ranged from 28 kg to 70 kg.

The experimental setting was as described in Zupan et al. (2012). Pigs were tested in five blocks of six pigs per block, with two trios per block. Briefly, behavioural and physiological responses by the measures of HR and HRV were recorded during the NOT and the NAT. Animals were

Table 1 - Definitions of behaviours observed in the novel object test (NOT) and novel arena test (NAT)

Behaviour	Definition
Lying (duration)	Lying or sitting without investigation of novel object
Standing (duration)	Standing without investigation of novel object
Walking (duration)	Locomotion of more than one step without investigation of novel object
Object investigation (duration, latency) ¹	Nosing, biting, licking, or manipulating novel object

The GENMOD procedure that fits generalized linear models was utilized for records of behaviour in the tests. The dependent variables were count values per animal (excretion frequency) or the duration of time in seconds (other recorded variables).

The GLM procedure was applied for each of the cardiac parameters separately. Pair-wise differences were tested by a multiple post hoc Tukey-Kramer test. Individuals were used as the subject for the repeated statement. The effect of body weight was tested in each model as a covariate but no significant effect was found.

Baseline data were used as covariate in the models. While in the NAT a significant influence of breed on HR was found, walking duration data were added as a covariate in the model to assess the effect of physical activity on HR. Activity could not explain variance in the data.

To investigate Spearman correlation coefficients, procedure CORR was used to assess the relationship between the HRV variables in all tested conditions. In the text, only Bonferroni-corrected statistical significant values are reported (P<0.05, B: P<0.004).

Table 2 - Total number of pigs used for particular statistical analysis

	Test	Fixed effect	N	Level	n
Behaviour	Novel object test	Sex	24	Gilt	15
				Barrow	9
		Breed		LY	14
				LYLD	10
	Novel arena test	Sex	27	Gilt	17
				Barrow	10
		Breed		LY	14
				LYLD	13
Heart rate variability (baseline)		Sex	13	Gilt	9
				Barrow	4
		Breed		LY	9
				LYLD	4
Heart rate variability (baseline/testing)	Novel object test	Sex	14	Gilt	8
				Barrow	6
		Breed		LY	9
				LYLD	5
	Novel arena test	Sex	11	Gilt	8
				Barrow	3
		Breed		LY	7
				LYLD	4

nd standing plus longer duration object but shorter duration of rows ($P<0.05$). The behavioural responses in the NAT test did not differ between sexes. Analyzing the breed differences during the exposure to the NOT, the duration of lying, standing, and walking was longer in LY pigs, while the duration of contact with a

(Table 4).

The combined baseline values of the cardiac activity parameters differed between breeds but not between gilts and barrows (Table 5). In detail, LY pigs had a significantly lower baseline HR and LF with a higher RMSSD and HF than LYLD pigs ($P<0.05$). When looking at the correlations between parameters of HRV, patterns

Table 3 - Effect of sex and genotype on behaviour during the novel object test

Behaviour	Fixed effect	Level	Mean $\pm$ SE	χ^2 test	P-value
Lying (duration)	Sex	Gilt	164.5 $\pm$ 46.2	40.59 ^{df=1,21}	<0.0001
		Barrow	149.1 $\pm$ 60.9		
	Genotype	LY	216.4 $\pm$ 47.8	803.19 ^{df=1,21}	<0.0001
		LYLD	77.9 $\pm$ 58.0		
Standing (duration)	Sex	Gilt	45.0 $\pm$ 15.3	6.48 ^{df=1,21}	<0.05
		Barrow	42.6 $\pm$ 20.2		
	Genotype	LY	58.7 $\pm$ 15.8	181.96 ^{df=1,21}	<0.0001
		LYLD	23.6 $\pm$ 19.3		
Walking (duration)	Sex	Gilt	13.8 $\pm$ 6.8	18.61 ^{df=1,21}	<0.0001
		Barrow	22.8 $\pm$ 9.0		
	Genotype	LY	21.4 $\pm$ 7.1	29.14 ^{df=1,21}	<0.0001
		LYLD	11.3 $\pm$ 8.6		
Latency to contact a novel object	Sex	Gilt	4.2 $\pm$ 1.9	0.21 ^{df=1,17}	>0.10
		Barrow	3.6 $\pm$ 2.7		
	Genotype	LY	3.6 $\pm$ 2.1	0.42 ^{df=1,17}	>0.10
		LYLD	4.3 $\pm$ 2.5		
Contact (duration)	Sex	Gilt	183.3 $\pm$ 37.4	14.53 ^{df=1,21}	<0.001
		Barrow	172.4 $\pm$ 49.3		
	Genotype	LY	79.8 $\pm$ 38.7	1864.25 ^{df=1,21}	<0.0001
		LYLD	318.5 $\pm$ 46.9		

SE - standard error.

LY - Landrace $\times$ Yorkshire; LYLD - Landrace/Yorkshire $\times$ Landrace/Duroc.

The data were modeled for a particular behavioural variable using phenotype, breed, and sex as tested effects. Results are presented in seconds and standard errors.

Table 4 - Effect of sex and genotype on behaviour during the novel arena test

Behaviour	Fixed effect	Level	Mean $\pm$ SE	χ^2 test	P-value
Lying (duration)	Sex	Gilt	56.1 $\pm$ 22.2	0.12 ^{df=1,24}	>0.10
		Barrow	54.4 $\pm$ 29.6		
	Genotype	LY	55.9 $\pm$ 24.4	0.38 ^{df=1,24}	>0.10
		LYLD	55.1 $\pm$ 26.1		
Standing (duration)	Sex	Gilt	254.8 $\pm$ 18.3	0.16 ^{df=1,24}	>0.10
		Barrow	255.3 $\pm$ 24.3		
	Genotype	LY	266.2 $\pm$ 20.1	14.52 ^{df=1,24}	<0.0001
		LYLD	242.9 $\pm$ 21.4		
Walking (duration)	Sex	Gilt	169.1 $\pm$ 9.9	0.72 ^{df=1,24}	>0.10
		Barrow	170.3 $\pm$ 13.3		
	Genotype	LY	157.9 $\pm$ 10.9	23.68 ^{df=1,24}	<0.0001
		LYLD	182.0 $\pm$ 11.7		

SE - standard error.

		LYLD	165.6±5.7		
	Sex	Gilt	22.7±4.5	0.40 ^{df=1,11}	>0.10
		Barrow	18.6±5.1		
	Genotype	LY	24.1±3.8	5.05 ^{df=1,11}	<0.05
		LYLD	15.5±5.7		
log10LF	Sex	Gilt	1181.2±214.8	0.04 ^{df=1,11}	>0.10
		Barrow	630.2±322.2		
	Genotype	LY	1292.6±198.4	6.84 ^{df=1,11}	<0.05
		LYLD	379.5±297.5		
log10HF	Sex	Gilt	136.7±39.0	0.45 ^{df=1,11}	>0.10
		Barrow	73.5±43.9		
	Genotype	LY	149.9±33.0	6.84 ^{df=1,11}	<0.05
		LYLD	43.8±49.3		
LF/HF ¹	Sex	Gilt	1070.5±140.9	0.01 ^{df=2,10}	>0.10
		Barrow	1146.4±158.4		
	Genotype	LY	1150.9±119.3	0.12 ^{df=2,10}	>0.10
		LYLD	965.4±177.9		

SE - standard error.

LY - Landrace × Yorkshire; LYLD - Landrace/Yorkshire × Landrace/Duroc.

RMSSD - root mean square of successive R-R intervals; log10LF - log transformed values tested for power of the low-frequency band; log10HF - log transformed values tested for power of the high-frequency band; LF - power of the low-frequency band; HF - power of the high-frequency band.

¹ Calculated on the original values of LF and HF.

Table 6 - Mean heart rate (HR) and heart rate variability parameters of pigs of different sexes and genotypes while pre-tested (baseline) and tested in the novel object test

Parameter	Fixed effect	Level	Mean ± SE		F-test	P-value
			Baseline	Testing		
log10HR (bpm)	Sex	Gilt	152.6±7.9	142.5±7.7	0.03 ^{df=5,8}	>0.10
		Barrow	152.7±8.9	160.5±8.6		
	Genotype	LY	146.8±6.9	143.6±6.8	1.03 ^{df=5,8}	>0.10
		LYLD	163.2±9.6	162.2±9.3		
RMSSD (ms)	Sex	Gilt	13.6±2.7	13.3±2.4	0.07 ^{df=5,8}	>0.10
		Barrow	9.3±3.0	8.6±2.7		
	Genotype	LY	13.7±2.4	12.2±2.1	0.78 ^{df=5,8}	>0.10
		LYLD	8.2±3.3	9.6±2.9		
LF	Sex	Gilt	860.9±249.3	1238.1±444.6	0.01 ^{df=5,8}	>0.10
		Barrow	620.7±287.9	585.5±516.6		
	Genotype	LY	160.7±56.1	1331.1±412.9	0.02 ^{df=5,8}	>0.10
		LYLD	31.1±75.3	217.7±555.7		
HF	Sex	Gilt	146.4±67.4	137.5±54.9	0.00 ^{df=5,8}	>0.10
		Barrow	71.7±75.8	57.7±61.8		
	Genotype	LY	160.7±59.5	137.8±48.5	0.25 ^{df=5,8}	>0.10
		LYLD	13.1±82.2	41.1±66.9		
LF/HF ¹	Sex	Gilt	1078.8±181.8	625.8±109.2	0.12 ^{df=5,8}	>0.10
		Barrow	1143.3±204.4	1022.4±123.3		
	Genotype	LY	1096.7±160.4	750.6±96.8	0.09 ^{df=5,8}	>0.10
		LYLD	1123.9±221.5	877.1±133.6		

SE - standard error.

LY - Landrace × Yorkshire; LYLD - Landrace/Yorkshire × Landrace/Duroc.

log10HR - log transformed values tested for heart rate; RMSSD - root mean square of successive R-R intervals; LF - power of the low-frequency band; HF - power of the high-frequency band.

¹ Calculated on the original values of LF and HF.

	Sex	Gilt	150.6±8.8	159.5±6.5	1.94 ^{df=5,5}	>0.10
		Barrow	151.7±13.2	155.7±9.7		
	Genotype	LY	149.9±9.3	152.4±6.8a	10.18 ^{df=5,5}	<0.05
		LYLD	152.8±13.5	169.0±9.9b		
RMSSD (ms)	Sex	Gilt	13.9±2.1	13.7±2.0	0.02 ^{df=5,5}	>0.10
		Barrow	7.8±3.2	11.4±2.9		
	Genotype	LY	12.2±2.2	13.7±2.1	0.48 ^{df=5,5}	>0.10
		LYLD	12.3±3.3	11.9±3.1		
log10LF	Sex	Gilt	1151.2±346.8	1515.0±607.7	0.02 ^{df=5,5}	>0.10
		Barrow	450.6±561.9	1228.4±984.5		
	Genotype	LY	1176.2±387.8	1706.2±679.4	0.02 ^{df=5,5}	>0.10
		LYLD	582.0±505.6	965.5±885.8		
log10HF	Sex	Gilt	128.1±32.3	176.3±68.5	0.05 ^{df=5,5}	>0.10
		Barrow	52.7±48.2	139.1±102.2		
	Genotype	LY	114.9±33.8	209.6±71.7	0.04 ^{df=5,5}	>0.10
		LYLD	94.7±49.5	90.1±104.9		
LF/HF ¹	Sex	Gilt	890.9±119.9	960.3±127.8a	10.75 ^{df=3,7}	<0.05
		Barrow	857.4±179.0	1013.5±190b		
	Genotype	LY	995.6±125.6	885.9±133.8	2.04 ^{df=3,7}	>0.10
		LYLD	682.4±183.8	1130.3±195.7		

SE - standard error.

LY - Landrace × Yorkshire; LYLD - Landrace/Yorkshire × Landrace/Duroc.

RMSSD - root mean square of successive R-R intervals; log10LF - log transformed values tested for power of the low-frequency band; log10HF - log transformed values tested for power of the high-frequency band; LF - power of the low-frequency band; HF - power of the high-frequency band.

¹ Calculated on the original values of LF and HF.

of Spearman correlation showed that in LY pigs, LF was positively correlated with HF ($r_s > 0.5$) and negatively with HR ($r_s < -0.7$). Heart rate was also negatively correlated with HF ($r_s < -0.7$), while a positive correlation was found ($r_s > 0.6$) between RMSSD and HF. In LYLD pigs, a strong positive correlation was found between RMSSD and LF/HF ratio ($r_s > 0.8$). Furthermore, in gilts, HR was strongly negatively correlated with LF and HF (both $r_s < -0.8$) and LF was positively correlated with HF ($r_s > 0.6$).

The pigs did not develop significant different autonomic responses under NOT (Table 6). Regarding the correlation analysis, it was revealed that in gilts during the test period, LF was strongly positively correlated with both RMSSD and HF, and HF was positively correlated with RMSSD (overall $r_s > 0.9$). In both breeds, HF was found to be strongly positively correlated with RMSSD ($r_s > 0.9$) and LF ($r_s > 0.8$), while in LYLD pigs, an additional positive correlation was found between RMSSD with LF ($r_s = +1$). In the NAT (Table 7), the LF/HF ratio was affected by sex. Barrows showed a significantly higher increase in LF/HF ratio than gilts did ($P < 0.05$). During the test period, the ratio was found to be negatively correlated with HF ($r_s = -1$) in barrows, while RMSSD was positively

positively correlated with HF ($r > 0.9$) and negatively with HR ($r_s < -0.8$). We found a significant difference between breeds in HR during test period. A higher HR was observed in LYLD pigs than in LY pigs ($P < 0.05$) and the results were not affected by walking activity of the animals (F-value = 0.17; $df = 5, 5$; $P > 0.10$). Pigs did not differ in other HRV parameters. With LYLD pigs, the correlation coefficient revealed that during social isolation RMSSD was strongly positively correlated with both LF and HF ($r_s = +1$).

Discussion

The most important finding of this study is the difference in basal cardiac activity across breeds but not sexes. Landrace × Yorkshire (LY) pigs had lower HR with higher RMSSD, LF, and HF when compared with Landrace/Yorkshire × Landrace/Duroc (LYLD) pigs. In addition, sex and breed affected behavioural responses during exposure to the NOT, whilst during exposure to the NAT only breed had an effect. Considering the cardiac responses in the NAT, HR increased more from baseline to testing in LY pigs than in LYLD pigs. Sympathovagal dominance also increased more from baseline to test period in barrows (increased

eline levels of vagal tone could y individuals, while low vagal llostasis (Korte et al., 2007). By showing higher RMSSD and HF, LY pigs may have had a less suppressed vagal activity than LYLD pigs and may thus be better adapted to different challenges.

During exposure to a novel-challenging situation in the NOT, gilts were witnessed as more exploratory and attentive than barrows, which supports our tested hypothesis. They showed longer duration of time spent actively nosing, biting, licking, or manipulating the novel object and longer duration of standing. It can be argued that showing longer duration of lying, during which animals often spent time manipulating the litter and most likely having their attention directed towards the novel object, is an additional evidence of gilts being more explorative and paying more attention to the novel object that they can manipulate. The behavioural changes observed across sexes together with the lower duration of walking may suggest that gilts may be less fearful than barrows (Forkman et al., 2007). It has been often reported that changes in behaviour follow the changes in physiology, so we would have expected differences in cardiac activity between sexes, but no differences were found. Since a lower motor activity level found in our gilts can be, according to the classical models of motivation (Hughes and Duncan, 1988), seen as a response to a perceived lack of control over the environment, which is stress-induced, we find this less likely when taking into account both behavioural and physiological measurements.

The reverse picture to that seen in the NOT emerged in the NAT, in which gilts showed similar behavioural responses to the social isolation in the novel environment but different cardiac activity compared with barrows. In the NAT, the increased LF/HF ratio values from baseline to testing in both gilts and barrows are neuro-physiological evidence of comparable mechanisms underlying behavioural responses when exposed to a novel situation. However, since gilts had a significantly higher LF/HF ratio than the barrows, it is possible to argue that the mechanism was regulated on a different level. The fact that we found no changes in RMSSD and HF, in parameters that were previously used as indicators of vagal activity in cattle (Després et al., 2002; Kovács et al., 2014) and pigs

of effect with negative valence (Yeates and Main, 2008), then the degree of aversiveness of the stimuli in the NAT was greater in barrows than in gilts as indicated by the greater increase in LF/HF. Owing to this and other reports (Porges, 1995; Friedman and Taylor, 1998), it is reasonable to think that barrows showed higher stress susceptibility during exposure to stressors in the NAT for having a higher LF/HF compared to gilts.

Our behavioural results indicate that environmental stimuli in the NOT (but not in the NAT) induced different behavioural responses in gilts when compared with barrows, which goes in line with a review by Forkman et al. (2007), who documented that specific responses of an individual depend on the nature of the novelty-related stimulus due to fluctuations in environmental or motivational variables. Furthermore, our results may also indicate different concepts of optimal exploratory strategy in gilts compared with barrows. As found for the behavioural responses, cardiac responses were not consistent over the two fear tests. While autonomic regulation of the heart varied in pigs of different sexes in the NAT, no such difference was found in the NOT. According to our results, we cannot fully confirm the previous findings showing that stress affects the sexes differently (Kritas and Morrison, 2007), but it is suggested that the response depends on the type of stressor presented.

Analyzing breed, LYLD pigs in comparison with LY pigs had a longer attention towards the novel object and a lower level of avoidance or fearfulness as indicated by showing longer duration of contact with a novel object in the NOT. Showing more lying, standing, and walking as indicators of less goal attention, a lower excitement for a novel object is furthermore confirmed in LY pigs. In the NAT, the exposure to novel stimuli elicited alterations in behavioural responses in LY pigs showing shorter latency to excrete, more standing, and less walking compared with LYLD pigs. Following the argument that a decreased motor activity level is an indicator of a perception of decreased control over the environment, which is stress-induced, while an increased level reflects an attempt of the animal to extricate itself from its surroundings prior to the onset of the learned helplessness (Hughes and Duncan, 1988), LY pigs seem to be more stressed in the NAT while less in the NOT when compared with LYLD pigs. From the measurements

g the exposure to stressors in the act of the breed on the magnitude sent the net interaction between vagal and sympathetic regulation (Camm et al., 1996). Increased HR is usually accompanied by vagal withdrawal (Sato et al., 2004) and this was numerically confirmed in our study by the lower RMSSD and HF from baseline to testing. Similar HR dynamics by fear-related response of HR and sympathetic activation has been described before (Stiedl et al., 2009). An alteration in HR but not in LF/HF ratio was reported by Visser et al. (2002) when investigating a reaction to psychological stress in horses. Based on our results and if we consider the changes of the sympathetic system to represent the arousal dimension of effect proposed by Yeates and Main (2008), we propose that LYLD pigs were more aroused by the stimuli in a socially isolated novel environment when compared with LY pigs.

In the NOT, both sex and breed had a significant impact on pig behaviour, with an exception of the latency to approach a novel object, while in the NAT it was only breed. A display of similar latency can be explained by the fact that pigs are in general curious animals that like to approach and explore new objects in their familiar environment. Interestingly, in the NOT, the results reflected neither sex nor breed influence on physiological responses in pigs, but they did in the NAT. It seems that in a novel situation that provokes an attentive response associated with excitement or fear, pigs have a similar attentive state regardless of their genetic background. This leads us to think that the exposure to a novel object may not be stressful for pigs in our study or that stimuli in the NAT were more stressful compared with the stimuli in the NOT, which supports our hypothesis. Another explanation for the obtained results is a small sample size used that may have prevented detection of existing genetic influence on autonomic regulation in the NOT.

A strong positive correlation was found between LF on the one hand and RMSSD and HF on the other hand during baseline conditions in gilts and the two investigated breeds. The same correlations were witnessed during test period in the NOT. In the NAT, the positive correlation was found in both sexes and in LYLD pigs. Finding these correlations emphasizes the strong association of LF with vagal activity in pigs from pens with tail biting outbreaks. The same

environment found in the current study led to an expression of emotion, possibly, fear, which was accompanied by elevated sympathetic tone. Although previous experience (i.e., phenotypic effects) could be problematic for the validity of baseline and reactive measures and present a confounding influence that cannot be excluded from the considerations here, our results give evidence that the sympathetic tone is more involved as a mediator of stress to the heart than vagal tone.

Conclusions

Pigs display complex behavioural and physiological interactions that characterize responses to social isolation in familiar and novel environments with differences relating to genetic background. Barrows seem to be less explorative than gilts and to have a higher level of fearfulness. When barrows are socially isolated in a novel environment, they have a higher low/high frequency band power ratio. Landrace/Yorkshire $\times$ Landrace/Duroc pigs are animals that perform more manipulative behaviours when occupied with a novel object, which may indicate a lower level of fearfulness than Landrace $\times$ Yorkshire pigs. By showing higher heart rate responses, Landrace/Yorkshire $\times$ Landrace/Duroc pigs also seem to get more aroused by novel stimuli in a socially isolated novel environment. Taken together, heart rate variability data can be an informative physiological measure related to presumed psychological processes in different sexes and breeds of pigs previously housed in pens in which tail biting occurs.

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